DressMe

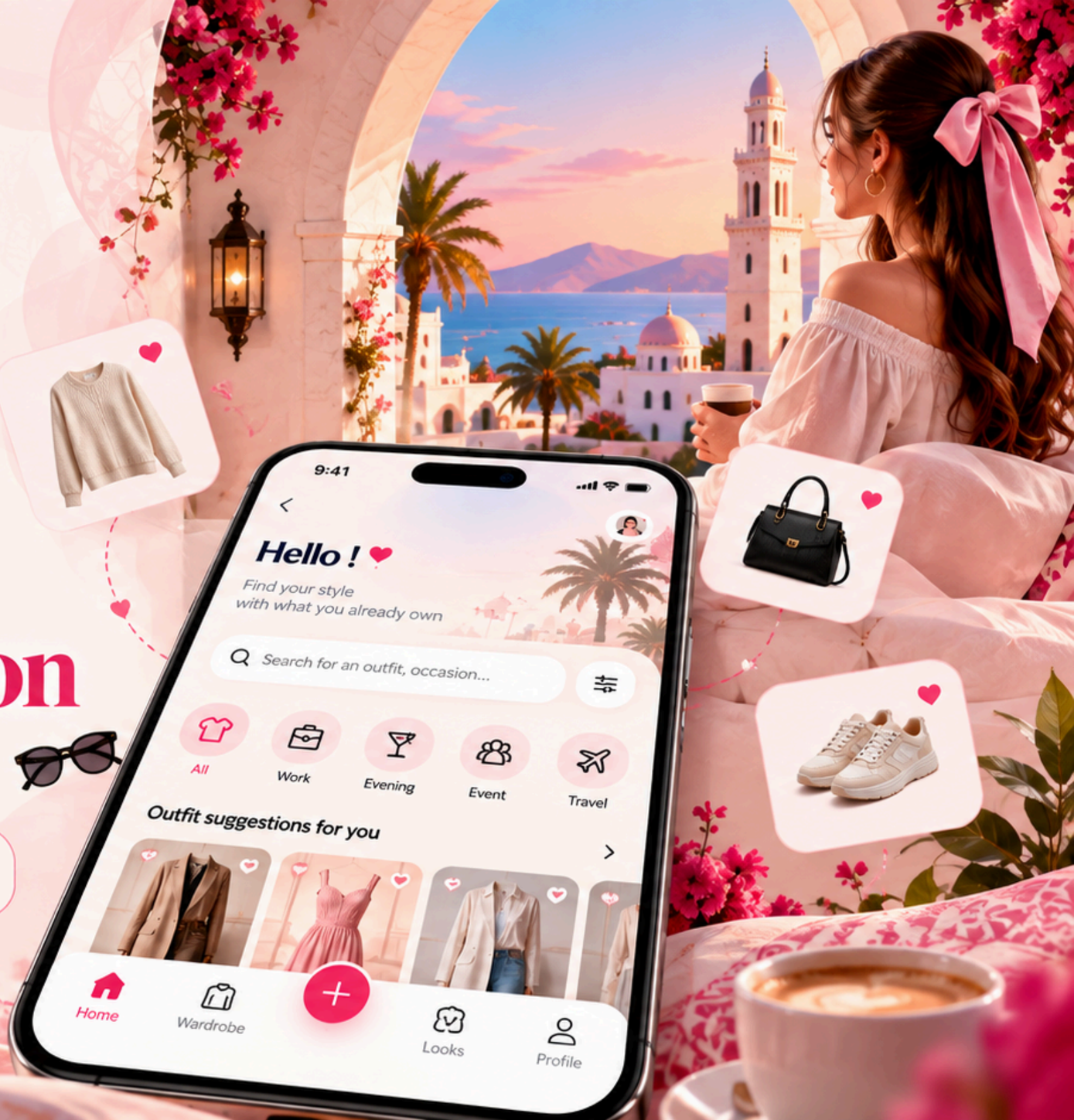
A SMARTER WARDROBE
FOR A BRIGHTER YOU

Project Presentation

Turning your wardrobe into opportunities

Advanced Data Science Project
AY 2026/2027

DressMe Team
Data Science & AI



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YOUR WARDROBE
YOUR INTELLIGENCE

Our Team

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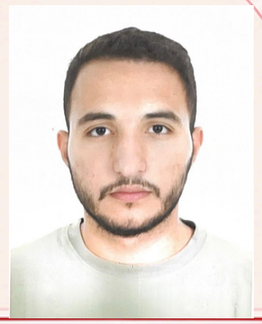

Hassan Zorkot

DressMe

VOTRE GARDE-ROBE,
PLUS INTELLIGENTE

TURNING
YOUR WARDROBE
INTO OPPORTUNITIES

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DressMe Overview

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A SMARTER WARDROBE FOR A BRIGHTER YOU

FROM CLOTHES YOU OWN
TO A SMARTER, BRIGHTER YOU

The Problem

- Underused wardrobe
- Uninspired purchases
- Lack of personalized guidance

Our Solution

Our Value Proposition

- Personalized outfit recommendations
- Helps you identify real gaps
- More conscious and optimized use of your wardrobe
- Access to matching Tunisian products

A smarter wardrobe for a brighter you

Ideation

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A wide exploration of ideas to reimagine a smarter fashion companion

Smart Wardrobe

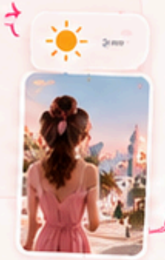
Digitize and understand your clothes automatically from photos.

Smart Outfit Generation

Create personalized outfits based on your style, context and existing wardrobe.

Context-Aware Styling

Adapt recommendations to your occasion, weather, location and mood.

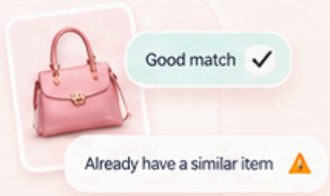


Style Learning & Profiling

Learn your style preferences over time from your feedback and choices.

Should I Buy This ?

Analyze a product and tell you if it fits your style, if it's redundant and if it's really worth buying.



Virtual Try-On

Virtually try outfits and visualize how they look on you.

Wardrobe Gap Detection

Identify missing pieces in your wardrobe to complete your style for key occasions.

Sustainable & Circular Fashion

Suggest outfit ideas for better reuse of existing clothes and facilitate donation or resale.

Local Fashion Matching

Find similar or complementary items from Tunisian brands and local stores.

Conversational Assistant

Get natural, personalized advice through a chat interface.

Wardrobe Organization

Help you organize, categorize and manage your clothes (e.g. seasonal, occasion, color).

Social & Community Inspiration

Discover outfits and trends from people with similar styles and share your looks.

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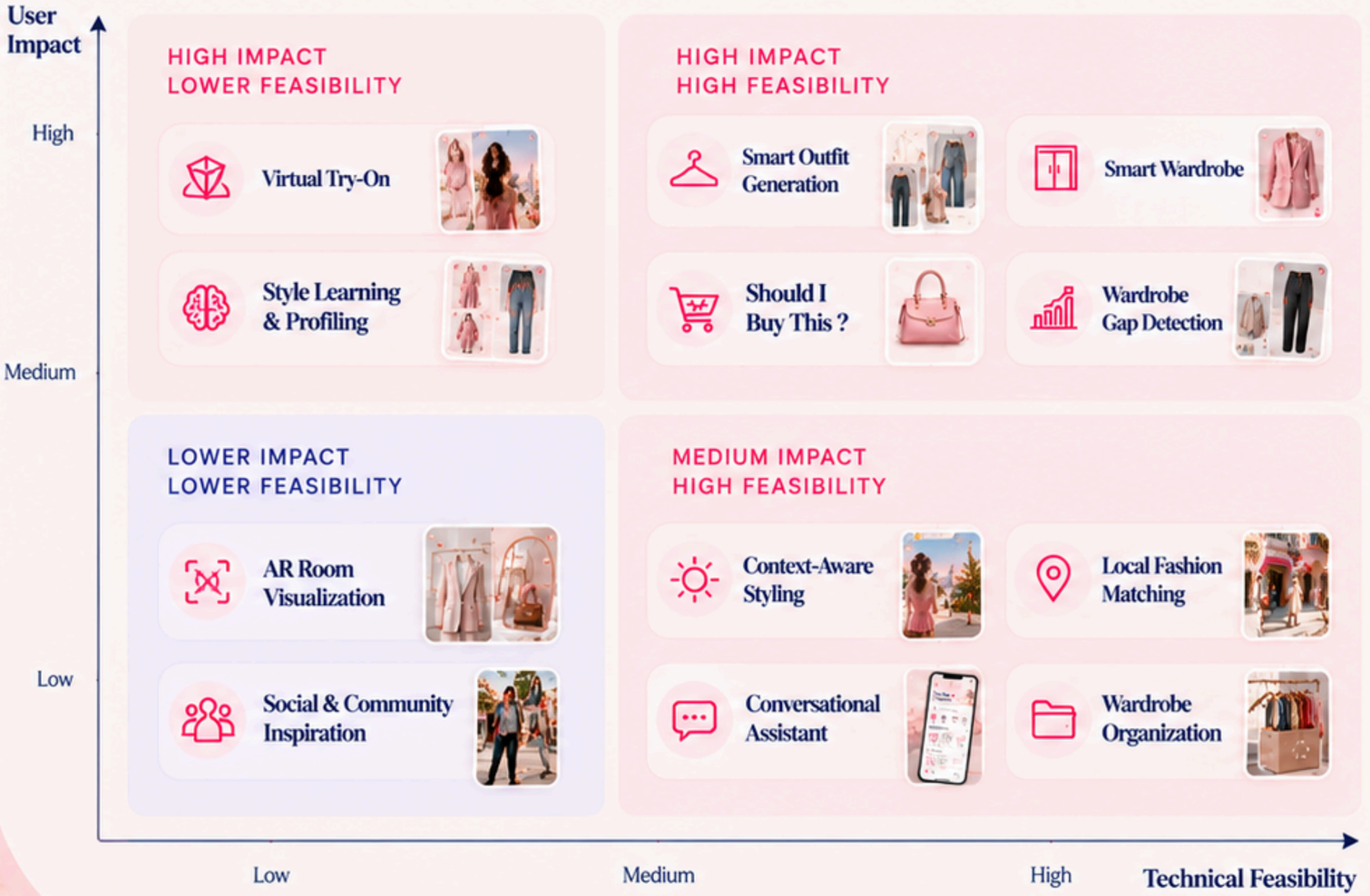
From Many Ideas to a Clear Direction

We explored a wide range of ideas, then prioritized the most valuable and feasible ones for our MVP.



Core / MVP

Selected for the prototype (S5)

1. Smart Wardrobe
2. Smart Outfit Generation
3. Should I Buy This ?
4. Wardrobe Gap Detection

Extensions

To be considered next

5. Context-Aware Styling
6. Local Fashion Matching
7. Conversational Assistant
8. Wardrobe Organization

Future Ideas

Long-term possibliities

9. Virtual Try-On
10. Style Learning & Profiling
11. AR Room Visualization
12. Social & Community Inspiration

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Selected Features & Data Science Objectives

Key features and the data science goals behind them.

Selected Features	Data Science Objectives	Expected Output
01 Smart Wardrobe	Clothing item recognition & classification	Pink Blazer — Jacket, Pink, Formal
02 Outfit Generation	Personalized outfit generation & ranking	
03 Should I Buy This ?	Compatibility and value prediction	Good match ✓
04 Fashion Matching	Similar item retrieval & recommendation	Similar items

05 | OUR DATA ECOSYSTEM

Our Data Ecosystem

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Multiple complementary data sources to power a personalized and local fashion experience.

Public Fashion Data

For fashion understanding and outfit compatibility

Polyvore

596 outfits analyzed

Outfit Compatibility

Tunisian Fashion Data

Local products for relevant and accessible recommendations

Hamadi Abid

715 Products

26 Categories

104 Colors

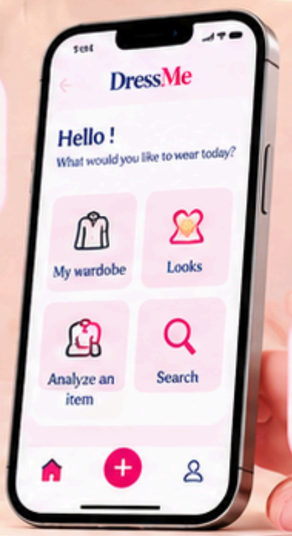
SCRAPED ✓

User & Context Data

For personalization and context-aware recommendations

Wardrobe (photos)

Occasion (work, casual, evening, ...)



Preferences (styles, colors, brands, budget)

Weather and season

Generated during app usage

Fashion Knowledge (Public Datasets) + **Local Catalog** (Tunisian Data) + **Personal Context** (User Data) → **Personalized Recommendations**

FROM RAW DATA
TO DRESSME INTELLIGENCE

From Raw Data to DressMe Intelligence

Data collection, cleaning and analysis to build a high-quality foundation for our AI system.

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Tunisian Fashion Catalog
Hamadi Abid

SCRAPED ✓

715 Products

26 Categories

104 Colors

Top Categories

T-shirts	112
Shoes	94
Accessories	71
Pants	63
Shirts	63
Shorts	55

Polyvore (Public Dataset)
Outfit compatibility & fashion knowledge

USED ✓

2,900 Items

596 Outfits analyzed

≈ 5 Items / outfit
(mean 4.87, median 5)

Outfit Size Distribution

Data Quality Check

131 / 2,900
non-fashion items detected

- 3 outfits fully non-fashion
- 88 outfits partially non-fashion
- 505 clean outfits

Raw Data — Scraped catalog & public datasets →
Explore — Data understanding and analysis →
Clean — Remove duplicates, filter non-fashion, standardize fields →
Structure — Organize and unify the data →
Ready for AI — Embeddings & fashion knowledge →
DressMe Intelligence — Personalized recommendations

DressMe MVP

A first look at our mobile prototype

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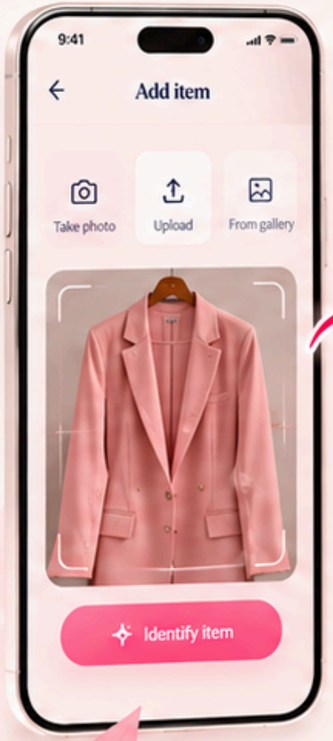
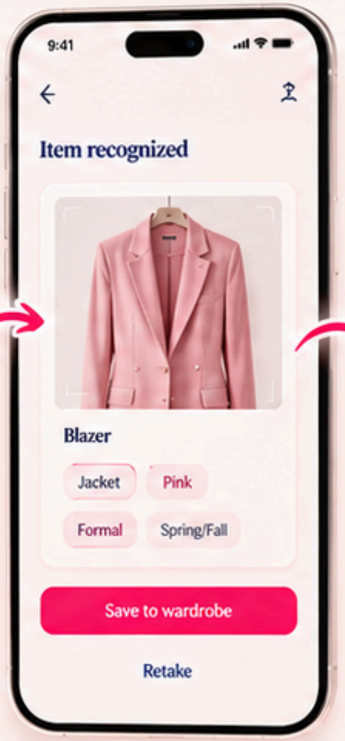
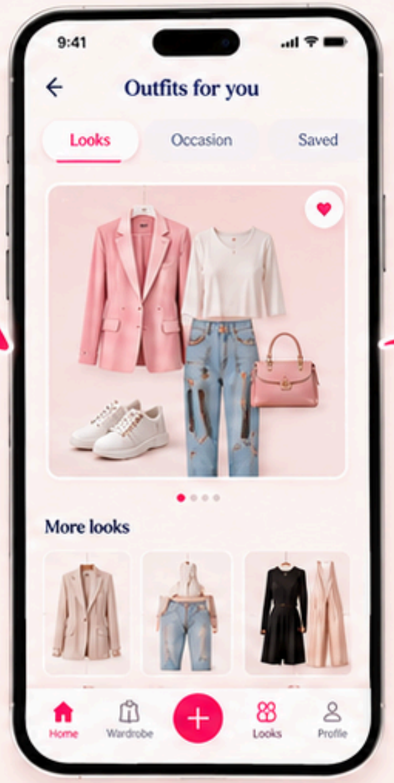
01 Smart Wardrobe
Upload → Recognize → Organize

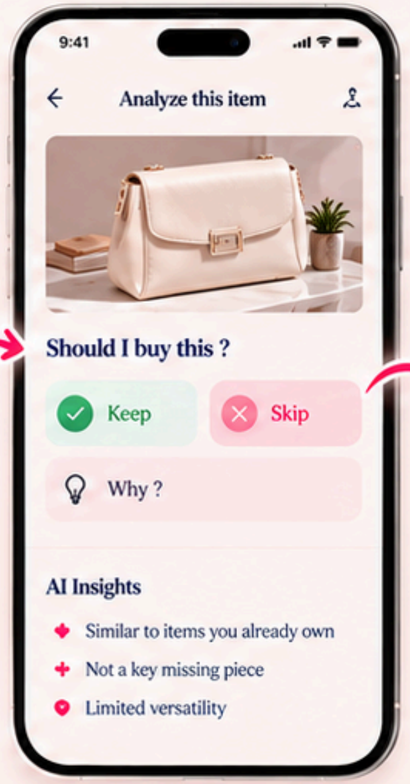




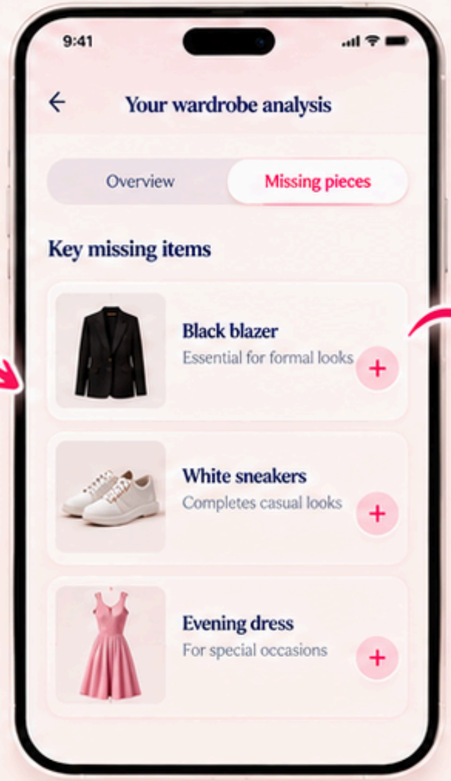
02 Outfit Generation
Personalized Looks



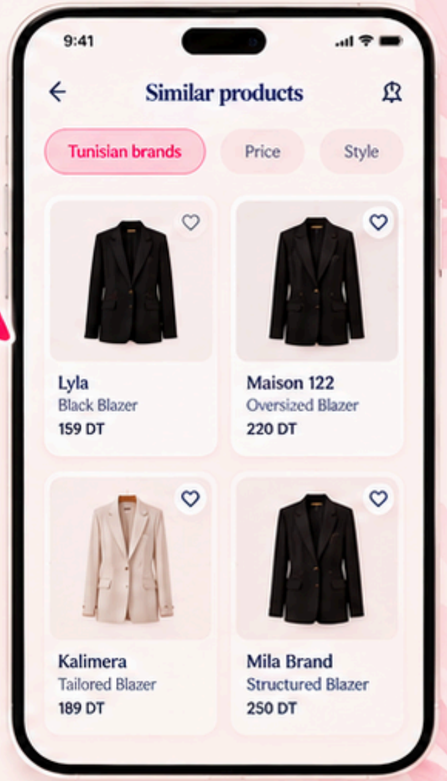
03 Should I Buy This ?
Keep / Skip / Why ?



04 Wardrobe Gap Detection
What is really missing ?



05 Local Fashion Matching
Real Gap Detected → Tunisian Tunisian Catalog



Use what you own first → Buy only when needed.

THE MVP VALIDATES DRESSME'S CORE INTELLIGENCE BEFORE ADDING ADVANCED FEATURES.

06 | PROTOTYPE (MVP)
USER JOURNEY

One Wardrobe. One Decision. One Journey.

From what I own to a complete look, only buying when needed.

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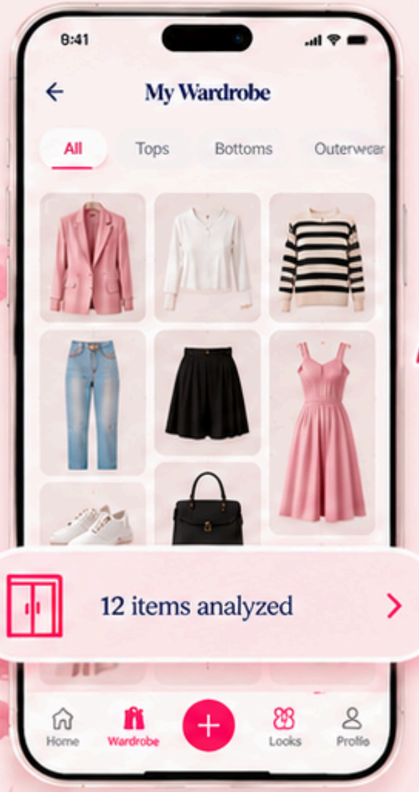
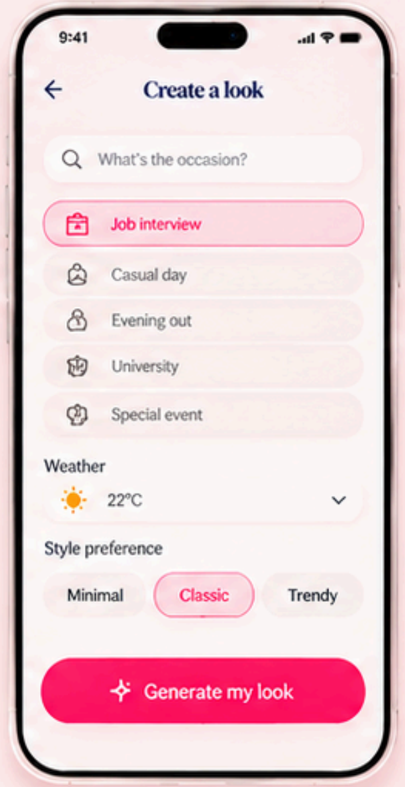
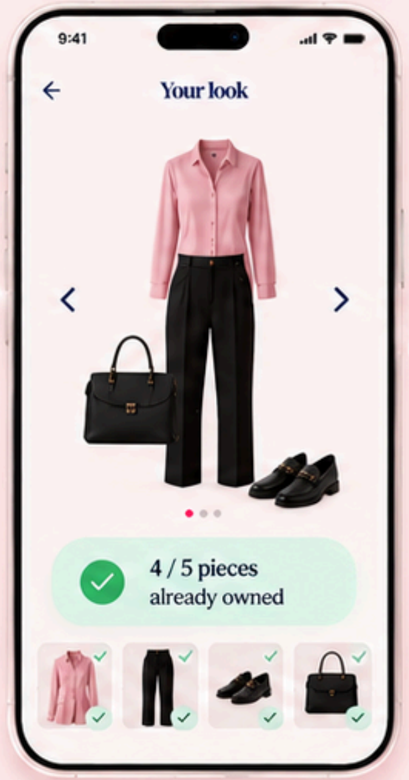
01 This is my wardrobe
Upload and organize your clothes



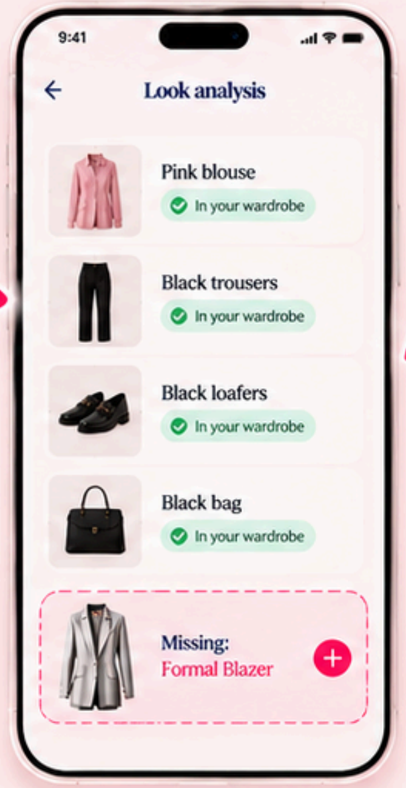
02 I need a look for...
Set your context and preferences



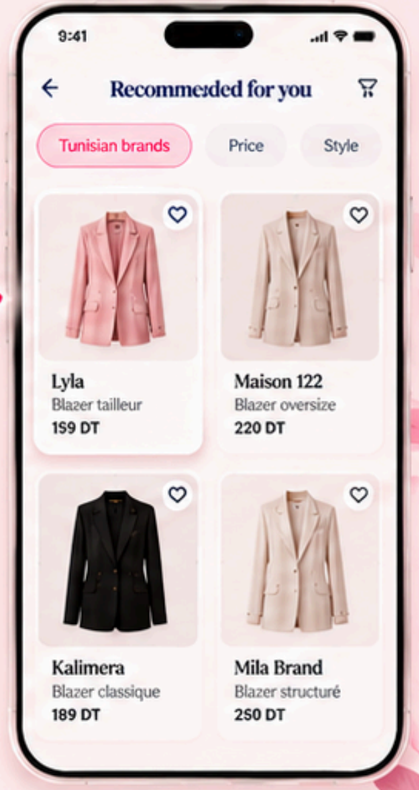
03 DressMe builds from what I already own
Get personalized outfit suggestions



04 One real gap detected
Identify what is missing



05 Only then, DressMe shops
Find similar items from Tunisian brands



Your Wardrobe — What you own
+
Your Style — Your preferences
+
Your Context — Occasion, weather...
→
Outfit — Personalized looks
→
Real Gap ? — Is something missing ?

- No → Wear what you own ♡
- Yes → Search Tunisian catalog

MVP Architecture

A modular and orchestrated architecture for a personalized mobile experience.

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User

- Add clothes (photos)
- Receive outfit recommendations
- Search for a new item
- Set your context (occasion, weather, budget...)

HTTP Request (JSON) →
← JSON Response

Mobile Application (iOS / Android)

React Native

HTTP Request (JSON) →
← JSON Response

Backend API

FastAPI

- Authentication (JWT)
- User management
- API endpoints
- Agent orchestration
- Data management

Orchestrator Agent

Analyzes the request, plans the tasks and coordinates the specialized agents.

Vision Agent

Analyzes clothing images

CNN / CLIP

Wardrobe Agent

Manages the user wardrobe

Python + SQL

Recommendation Agent

Generates personalized outfits

ML / Deep Learning

Purchase Agent

Analyzes a new item

CV + Similarity

LLM Agent (optional)

Generates natural and personalized explanations

GPT / LLaMA / Mistral

External Services (optional)

- Weather API
- Tunisian e-commerce API
- Social media APIs (Instagram, Pinterest)

Data Storage & Infrastructure

Relational Database (PostgreSQL)

- Users
- Wardrobe items
- Preferences
- History & feedback

Vector Database (Qdrant)

drant

- Image embeddings
- Similarity search
- Visual recommendations

Image Storage (Cloud Storage)

- Clothing photos
- Generated looks
- User files

AWS S3 / Google Cloud · aws

Deployment & Security

- JWT authentication
- Data encryption
- Docker deployment
- Monitoring & logs

docker · aws · GitHub

MVP Prototyping Plan

From data preparation to the integration of the MVP.

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01 Dataset Preparation

- Collect and clean data
- Remove duplicates and noise
- Organize metadata
- Define data schema

02 Tunisian Catalog Scraping

- Scrape local products (clothes, accessories....)
- Extract key information (price, brand, category, image....)
- Clean and structure data

03 Baseline Vision / Embeddings

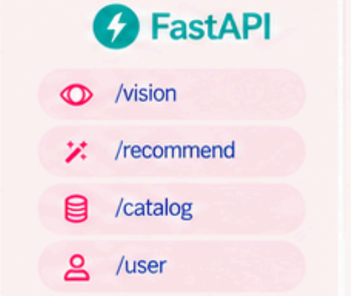
- Clothing recognition (CNN/CLIP)
- Extract visual embeddings
- Test on an initial dataset

04 Recommendation Agent

- Generate personalized outfits
- Detect wardrobe gaps
- Search for similar products
- Ranking and explanations

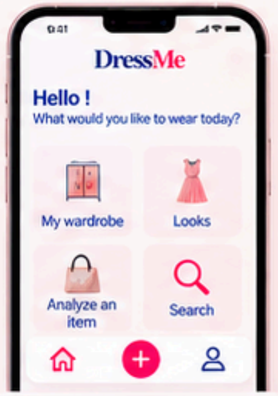
05 API / Backend

- Develop the API with FastAPI
- Manage user data
- Endpoints for each agent
- Connect to databases



06 Mobile Interface & Integration

- Develop the mobile app (React Native)
- Integrate agents and API
- End-to-end testing
- Iterative improvements



Week 1–2

- Data collection & cleaning
- Data schema definition

Week 2–3

- Tunisian catalog scraping
- Preprocessing & storage

Week 3–4

- Baseline vision implementation
- Embedding extraction

Week 4–5

- Recommendation agent development
- Initial testing & evaluation

Week 5–6

- API / backend development
- Database connection

Week 6–7

- Mobile interface
- Full integration
- Testing & iterations

A progressive prototyping plan to validate each component before the integration of the MVP.

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Conclusion

DressMe makes fashion more personal, conscious and local.

Thank You

for your attention !

- A project built with passion
- By a motivated team
- For a smarter and more conscious fashion
- And a brighter Tunisia